THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS V — MATHEMATICS · MODEL PAPER 09 SOLVED

Learning-Level Evaluation Pattern (RSCERT) | Session 2026–27

Time: 2.30 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

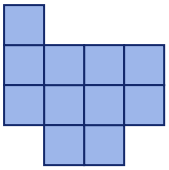
General Instructions:

1. This question paper is based on the learning areas of Class 5 Mathematics (Shape and Space, Number System, Operations, Measurement, Data Handling and Patterns).
2. Write your name and roll number in the space given on the question paper.
3. All questions are compulsory.
4. The marks for each question are shown against it.
5. Write your answers neatly and show your working wherever it is needed.

UNDERSTANDING SHAPE AND SPACE — Marks 21

1. The side of each square is 1 cm. The perimeter of the shaded figure is [1]

(a) 18 cm **(b)** 16 cm **(c)** 14 cm **(d)** 22 cm



Ans. (b) 16 cm

2. [6]

(i) Write the type of each angle shown: (a) ∠P (b) ∠Q (c) ∠R 3



∠P = 15° ∠Q = 175° ∠R = 270°

Ans. (a) acute (less than 90°) (b) obtuse (just less than 180°) (c) reflex (more than 180°)

(ii) (a) Through how many degrees does the hour hand turn from 12 to 5? (b) What type of angle is made by the hands of a clock at 3 o'clock? (c) How many straight angles make a complete turn? 3

Ans. (a) 5 × 30 = **150°** (b) right angle (c) 360 ÷ 180 = **2**

3. [6]

(i) Name the solid that looks like: (a) a brick (b) a funnel (c) the Great Pyramid of Egypt 3

Ans. (a) cuboid (b) cone (c) square pyramid

(ii) Use the table. (a) Which two solids have the same number of faces and vertices? (b) Which solid has the most vertices? (c) What is the total number of edges of a cube and a triangular prism together? 3

Solid	Faces	Edges	Vertices
Cube	6	12	8
Triangular pyramid	4	6	4
Square pyramid	5	8	5
Triangular prism	5	9	6

Ans. (a) **Triangular pyramid** (4, 4) and **square pyramid** (5, 5) (b) **Cube** (8) (c) 12 + 9 = **21**

4. The plan of Hari's house in Kota is given. **[8]**



(a) Find the perimeter of the Living Room. **2**

Ans. 2 × (7 + 4) = **22 m**

(b) Find the area of the Store. **2**

Ans. 3 × 4 = **12 sq m**

(c) In which direction (with respect to the Living Room) is the Store? **2**

Ans. East (it is on the right side; north is at the top)

(d) A carpet costing ₹ 85 per sq m is laid in the Bedroom. Find the cost. **2**

Ans. Area = 7 × 3 = 21 sq m; cost = 21 × 85 = **₹ 1,785**

NUMBER SYSTEM — Marks 9

5. In the number 6,08,347 the digit in the ten-thousands place is **[1]**

(a) 6 **(b)** 8 **(c)** 0 **(d)** 3

Ans. (c) 0

6. The smallest 6-digit number that can be made from the digits 7, 0, 4, 2, 9, 6 (each used once) is **[1]**

(a) 2,04,679 **(b)** 24,679 **(c)** 2,04,769 **(d)** 2,46,790

Ans. (a) 2,04,679

7. 705009 is read as **[1]**

(a) seven lakh fifty thousand nine **(b)** seventy lakh five thousand nine

(c) seven lakh five thousand ninety **(d)** seven lakh five thousand nine

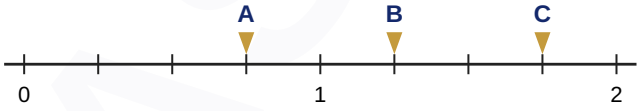
Ans. (d) seven lakh five thousand nine — 7,05,009

8. **[6]**

(i) A potter of Kota made three pots weighing 1.25 kg, 2.5 kg and 0.75 kg. (a) What is the total weight? (b) How much heavier is the heaviest pot than the lightest? (c) Is the total more or less than 5 kg? **3**

Ans. (a) 1.25 + 2.50 + 0.75 = **4.50 kg** (b) 2.50 − 0.75 = **1.75 kg** (c) less (4.5 kg < 5 kg)

(ii) The line from 0 to 2 is divided into 8 equal parts. Write the fraction shown by A, B and C (write B and C as mixed numbers). **3**



Ans. A = 3/4 B = 5/4 = 1 1/4 C = 7/4 = 1 3/4

OPERATIONS — Marks 25

9. Which of these numbers is divisible by 9? **[1]**

(a) 4,921 **(b)** 4,925 **(c)** 4,923 **(d)** 4,927

Ans. (c) 4,923 — 4 + 9 + 2 + 3 = 18, a multiple of 9

10. A school in Jodhpur buys 28 benches at ₹ 650 each. The shopkeeper gives a discount of ₹ 1,200 on the whole bill. How much money does the school pay? **[3]**

Ans. Cost of benches = 28 × 650 = ₹ 18,200
After discount = 18200 − 1200 = **₹ 17,000**

11. Divide 9,876 by 13. Write the quotient and the remainder, and check the answer by multiplication. **[3]**

Ans. Quotient = **759**, remainder = **9**.
Check: 13 × 759 = 9867; 9867 + 9 = 9876. ✓

12. [6]

(i) Ramlal sells 35 litres of milk every day at ₹ 52 per litre. (a) How much does he earn in a day? (b) How much does he earn in 30 days? (c) His monthly cost of cattle feed is ₹ 18,000. How much is left? 3

Ans. (a) 35 × 52 = **₹ 1,820** (b) 1,820 × 30 = **₹ 54,600** (c) 54,600 − 18,000 = **₹ 36,600**

(ii) Find using an easy method: (a) 52 × 98 (b) 36 × 25 (c) 4,000 − 1,997 3

Ans. (a) 52 × 100 − 52 × 2 = 5200 − 104 = **5,096** (b) 36 × 100 ÷ 4 = 3600 ÷ 4 = **900** (c) 4,000 − 2,000 + 3 = **2,003**

13. There are 20 lockers numbered 1 to 20. Student 1 touches every locker, Student 2 touches every 2nd locker (2, 4, 6, …), Student 3 touches every 3rd locker, and so on up to Student 20. (A student touches a locker if its number is a multiple of the student's number.) [6]

(1) (2) (3) (4) (5) (6) (7) (8) (9) (10)

(11) (12) (13) (14) (15) (16) (17) (18) (19) (20)

(i) (a) Which students touch locker 12? (b) How many students touch locker 16? (c) Which students touch locker 9? 3

Ans. (a) **1, 2, 3, 4, 6, 12** (the factors of 12) (b) factors of 16 are 1, 2, 4, 8, 16, so **5** students (c) **1, 3, 9**

(ii) (a) Which lockers are touched by exactly 2 students? (b) Which lockers does Student 6 touch? (c) Name the locker (up to 20) touched by both Student 4 and Student 6. 3

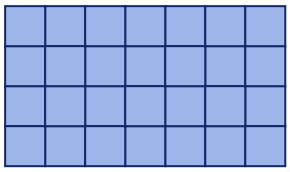
Ans. (a) **2, 3, 5, 7, 11, 13, 17, 19** (prime numbers) (b) **6, 12, 18** (c) **12** (LCM of 4 and 6)

14. [6]

(i) Solve the riddles: (a) I am a multiple of both 6 and 8, and I am between 40 and 60. (b) I am a 2-digit number less than 40. The sum of my digits is 9 and their product is 18. (c) I am a square number between 50 and 80. 3

Ans. (a) **48** (b) **36** (3 + 6 = 9, 3 × 6 = 18; 63 is more than 40) (c) **64** (8 × 8)

(ii) The floor of a room is covered by square tiles in 4 rows and 7 columns, as shown. (a) How many tiles are there? (b) Each tile has a side of 50 cm. Find the length and breadth of the floor in metres. (c) Find the area of the floor in sq m. 3



Ans. (a) 4 × 7 = **28** (b) length = 7 × 50 = 350 cm = **3.5 m**; breadth = 4 × 50 = 200 cm = **2 m** (c) 3.5 × 2 = **7 sq m**

MEASUREMENT — Marks 17

15. Fill in the blanks: (a) 7 km 250 m = _____ m (b) 2 days = _____ hours (c) 3 L 50 mL = _____ mL [3]

Ans. (a) 7250 (b) 48 (c) 3050

16. Write True or False: (a) 00:00 hours is midnight in the 24-hour clock. (b) 1 quintal is 10 kg. (c) The boiling point of water is 100°C. [3]

Ans. (a) True (b) False (1 quintal = 100 kg) (c) True

17. Ramesh started his work at 6:50 am and worked for 7 hours 25 minutes. At what time did he finish? Write the time in the 12-hour clock. [3]

Ans. 6:50 am + 7 h = 1:50 pm; 1:50 pm + 25 min = 2:15 pm.
He finished at **2:15 pm** (14:15).

18. The table shows the road distance (in km) between some cities. **[8]**

km	Jaipur	Ajmer	Kota	Bharatpur	Alwar
Jaipur	—	130	250	180	150
Ajmer	130	—	200	310	280
Kota	250	200	—	330	400
Bharatpur	180	310	330	—	110
Alwar	150	280	400	110	—

(a) A taxi goes from Jaipur to Ajmer and comes back to Jaipur. How many km does it run? **2**

Ans. 2 × 130 = **260 km**

(b) Which city is nearest to Alwar? **2**

Ans. Bharatpur (110 km)

(c) Which city is farthest from Kota? **2**

Ans. Alwar (400 km)

(d) Going from Alwar to Jaipur through Bharatpur is how many km longer than the direct road? **2**

Ans. 110 + 180 = 290 km; 290 − 150 = **140 km**

DATA HANDLING AND PATTERNS — Marks 8

19. The product of two odd numbers is always **[1]**

(a) an odd number **(b)** an even number

(c) a multiple of 4 **(d)** zero

Ans. (a) an odd number — e.g. 3 × 5 = 15, 7 × 9 = 63

20. Tiles are arranged in rectangles as shown. How many tiles will be needed for Fig. 6? **[1]**

(a) 30 **(b)** 36 **(c)** 49 **(d)** 42

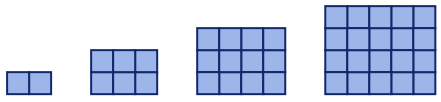


Fig. 1 Fig. 2 Fig. 3 Fig. 4

Ans. (d) 42 — 1×2, 2×3, 3×4, … → 6 × 7 = 42

21. A pictograph shows the vegetables sold in a shop. Each dot (●) stands for 10 kg. **[3]**

Day	Vegetables sold (● = 10 kg)
Mon	● ● ● ●
Tue	● ● ● ● ● ●
Wed	● ● ●
Thu	● ● ● ● ●

(a) How many kg of vegetables were sold on Thursday? **1**

Ans. 5 × 10 = 50 kg

(b) On which day was the least sold? **1**

Ans. Wednesday (3 baskets = 30 kg)

(c) How many more kg were sold on Tuesday than on Wednesday? **1**

Ans. 60 − 30 = 30 kg

22. Look at the pattern: 3, 4, 6, 9, 13, 18, ___, ___. (a) Write the number added at each step. (b) Write the next two numbers. (c) Which number is added to get the 9th number from the 8th? **[3]**

Ans. (a) +1, +2, +3, +4, +5, … (the added number goes up by 1 each time) (b) 18 + 6 = 24, 24 + 7 = 31 (c) the 8th number is 31, the 9th is 31 + 8 = 39, so **8** is added

END OF QUESTION PAPER — Total: 80 Marks